MAT 314 HW01

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Problem. 1.1.

(a) Is $GL_n(\mathbb{C})$ isomorphic to a subgroup of $GL_{2n}(\mathbb{R})$?

(b) Is $SO_2(\mathbb{C})$ a bounded subset of $C^{2\times 2}$?

Solution. For (a), consider the map

$$Z(a+bi) = \begin{pmatrix} a & -b \\ b & a \end{pmatrix},$$

which is an injective ring homomorphism $Z\colon \mathbb{C} \to \mathbb{R}^{2\times 2}$: that it is an injective homomorphism of the underlying additive groups is trivial to check, and it is well-behaved with multiplication because

$$(a+bi)(c+di) \mapsto \begin{pmatrix} a & -b \\ b & a \end{pmatrix}\begin{pmatrix} c & -d \\ d & c \end{pmatrix} = \begin{pmatrix} ac-bd & -ad-bc \\ ad+bc & ac-bd \end{pmatrix} = Z(ac-bd+(ad+bc)i).$$

Now let $A \in GL_n(F)$ for $F = \mathbb{C}$ or $\mathbb{R}$ have entry $A_{i,j}$ in row i, column j. Then, suppose that ϕ is a map sending $A = (A_{i,j}) \in GL_n(\mathbb{C})$ to $B = (B_{i,j}) \in GL_{2n}(\mathbb{R})$ with the rule

$$\begin{pmatrix} B_{2i-1,2j-1} & B_{2i-1,2j} \\ B_{2i,2j-1} & B_{2i,2j} \end{pmatrix} = Z(A_{i,j}).$$

In fact, it's actually a ring homomorphism $C^{1\times 1} \to \mathbb{R}^{2\times 2}$.

That is, if we abbreviate $Z(A_{i,j})$ as $Z_{i,j}$, then

$$\phi(A) = \begin{pmatrix} Z_{1,1} & Z_{1,2} & \cdots & Z_{1,n} \\ Z_{2,1} & Z_{2,2} & \cdots & Z_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ Z_{n,1} & Z_{n,2} & \cdots & Z_{n,n} \end{pmatrix}.$$

From here we can see that for $X, Y \in GL_n(\mathbb{C})$,

$$(\phi(X)\phi(Y))_{i,j} = \sum_{k=1}^{n} Z(X_{i,k})Z(Y_{k,j}) = Z\left(\sum_{k=1}^{n} X_{i,k}Y_{k,j}\right) = \phi(XY),$$

so ϕ is a homomorphism, and injectivity of ϕ follows from injectivity of Z. This establishes an isomorphism between $GL_n(\mathbb{C})$ and the subgroup of $GL_{2n}(\mathbb{R})$ corresponding to the image of ϕ.

For (b), we observe that

$$\left\{ \begin{pmatrix} x & i\sqrt{x^2-1} \\ -i\sqrt{x^2-1} & x \end{pmatrix} : x > 1 \right\}$$

is a subset of $SO_2(\mathbb{C})$, since the inverse of each of these matrices is its own transpose and their determinants are all 1, and of course since x can be arbitrarily large this shows that $SO_2(\mathbb{C})$ is unbounded. □

Problem. 3.4. The centralizer of $\mathbf{j}$ is just the subset of SU_2 whose $\mathbf{i}$ and $\mathbf{k}$ components are zero. Indeed,

$$a\mathbf{j} + b\mathbf{k} - c - d\mathbf{i} = (a + b\mathbf{i} + c\mathbf{j} + d\mathbf{k}) \cdot \mathbf{j} = \mathbf{j} \cdot (a + b\mathbf{i} + c\mathbf{j} + d\mathbf{k}) = a\mathbf{j} - b\mathbf{k} - c + d\mathbf{i}$$

implies that $b = -b = 0$ and $d = -d = 0$.

Problem. 5.1. Can the image of a one-parameter group in GL_n cross itself?

Solution. The answer is $\boxed{\text{yes}}$. Take $\phi\colon \mathbb{R}^+ \to GL_n$ to be the one-parameter group

$$\phi(t) = e^{tA}, \qquad A = \begin{pmatrix} 0 & -1 & 0 & 0 & \cdots & 0 \\ 1 & 0 & 0 & 0 & \cdots & 0 \\ 0 & 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & 1 \end{pmatrix}.$$

Then powers of A fix everything except for the top-left 2×2 submatrix, which is periodic under exponentiation by an integer with period 4. In particular

$$e^{tA} = \begin{pmatrix} \cos t & -\sin t & 0 & 0 & \cdots & 0 \\ \sin t & \cos t & 0 & 0 & \cdots & 0 \\ 0 & 0 & e & 0 & \cdots & 0 \\ 0 & 0 & 0 & e & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & e \end{pmatrix},$$

so the image of ϕ crosses itself, as for example $e^{0A} = e^{2\pi A}$. $\square$

Problem. 6.1. Verify the Jacobi identity for the bracket operation $[A, B] = AB - BA$.

Solution. First, observe that

$$[A, [B, C]] = A(BC - CB) - (BC - CB)A = ABC - ACB - BCA + CBA.$$

Similarly,

$$\begin{aligned} [B, [C, A]] &= BCA - BAC - CAB + ACB \\ [C, [A, B]] &= CAB - CBA - ABC + BAC \end{aligned}$$

and we can see easily that every term with a positive coefficient pairs off with its negative when adding the three brackets together. $\square$

Problem. 8.4.

(a) Write the polynomial equations that define the symplectic group.

(b) Show that the unitary group U_n can be defined by real polynomial equations in the real and imaginary parts of the matrix entries.

Solution. (a) Writing elements of GL_{2n} as blocks of $n \times n$ matrices

$$\begin{pmatrix} A & B \\ C & D \end{pmatrix},$$

we want

$$\begin{pmatrix} A & C \\ B & D \end{pmatrix} \begin{pmatrix} 0 & I \\ -I & 0 \end{pmatrix} \begin{pmatrix} A & B \\ C & D \end{pmatrix} = \begin{pmatrix} 0 & I \\ -I & 0 \end{pmatrix},$$

which gives the equations

$$\begin{aligned} A^t C - C^t A &= 0 \\ B^t D - D^t B &= 0 \\ A^t D - C^t B &= I, \end{aligned}$$

with the fourth equation $B^t C - D^t A = -I$ being redundant because it's obtained by taking the negative transpose of both sides of the third equation.

(b) Given a matrix in U_n with entries $z_{i,j}$ in row i, column j, we know the relations

$$\sum_{k=1}^{n} z_{i,k}\overline{z_{j,k}} = \begin{cases} 1, & i = j \\ 0, & i \neq j \end{cases}$$

are equivalent to the matrix being unitary, since this is exactly enough to characterize the matrix as being the inverse of its conjugate transpose. Splitting the $z_{i,j}$ into real and imaginary parts and using the above equations gives a collection of real polynomials that determine U_n. $\square$